

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Information Technology) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Information Technology) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTITC404 - DESIGN AND ANALYSIS OF ALGORITHMS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Which of the following is not one of the properties of algorithm?
 1. Multiplicity
 2. Ambiguity
 3. Finiteness
 4. Range of input
- 2 Which one of the following helps in calculating the largest amount of time taken for the completion of the algorithm?
 1. Theta notation
 2. Big-Oh notation
 3. Omega notation
 4. Alpha notation
- 3 The solution of $T(n) = 9T(n/3) + n^3$ recurrence relation using Master's method is _____
 1. $\Theta(n)$
 2. $\Theta(n^2)$
 3. $\Theta(n^3)$
 4. $\Theta(n \log n)$
- 4 The correct order of following functions by growth rate is: $n, n^2, n \log n, 2^n, n^3, n!$ is _____
 1. $n, n^2, n \log n, 2^n, n^3, n!$
 2. $n^3, n, n^2, n \log n, n!$
 3. $n, n \log n, n^2, n^3, 2^n, n!$
 4. $n^2, n \log n, 2^n, n^3, n!, n$

- 5 Time complexity of the Fibonacci sequence using Dynamic Programming (bottom-up) is _____
1. $O(2^n)$
 2. $O(n^2)$
 3. $O(n)$
 4. $O(n \log n)$
- 6 In the 0/1 Knapsack problem, we _____
1. Take full item or leave it
 2. Take fractions of item
 3. Always take heavier items
 4. Ignore weights
- 7 Multistage graph typically used for _____
1. Sorting data
 2. Finding the longest path
 3. Finding the shortest path through stages
 4. Balancing a binary tree
- 8 In Depth First Search (DFS), nodes are typically visited using _____
1. Queue
 2. Heap
 3. Stack
 4. Priority Queue
- 9 Which of the following algorithms is used to find an MST?
1. Dijkstra's Algorithm
 2. Bellman-Ford Algorithm
 3. Prim's Algorithm
 4. Floyd-Warshall Algorithm
- 10 _____ problems is commonly solved using backtracking.
1. Binary Search
 2. Merge Sort
 3. N-Queens Problem
 4. Dijkstra's Algorithm
- 11 What does the class P represent in computational theory?
1. Problems solved in exponential time
 2. Problems that cannot be solved
 3. Problems solvable in polynomial time
 4. Problems verifiable in polynomial time
- 12 _____ is the worst case time complexity of the quick sort algorithm.
1. $O(n)$
 2. $O(1)$
 3. $O(n \log n)$
 4. $O(n^2)$

Q2. Solve the following.

- A) Define Big O notation, Big Omega and Big Theta notation. Depict the same graphically and explain.

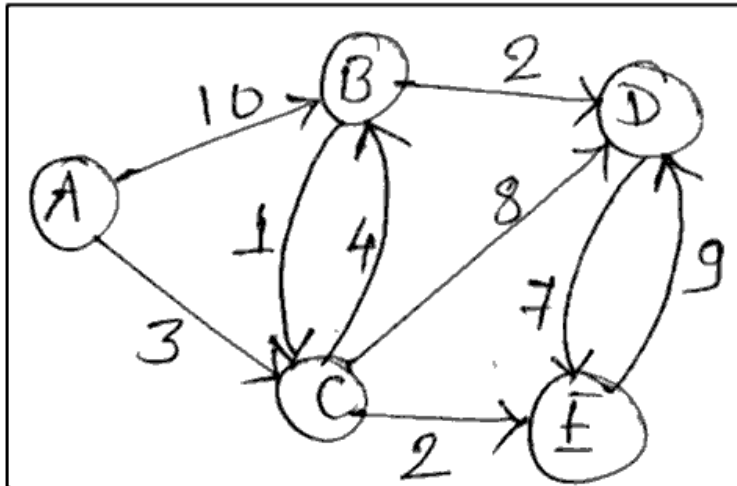
6

- B) Order the following functions by growth rate: n , n^2 , $n \log n$, 2^n , n^3 , $n!$. Also Solve following recurrence relation using Master's method: 6

1. $T(n) = 8T(n/2) + n^2$
2. $T(n) = 3T(n/3) + n^3$

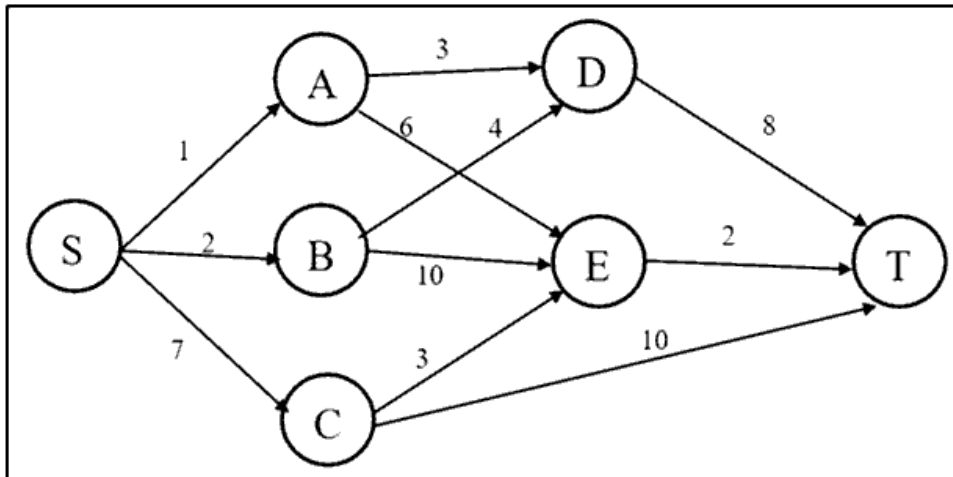
Q3. Solve the following.

- A) Formulate a greedy approach to obtain an optimal merge pattern to pair wise merge n sorted files. Write an algorithm to generate two-way merge tree. 6
- B) Compute single source shortest path using Dijkstra's algorithm. Assuming 'A' as source vertex. 6

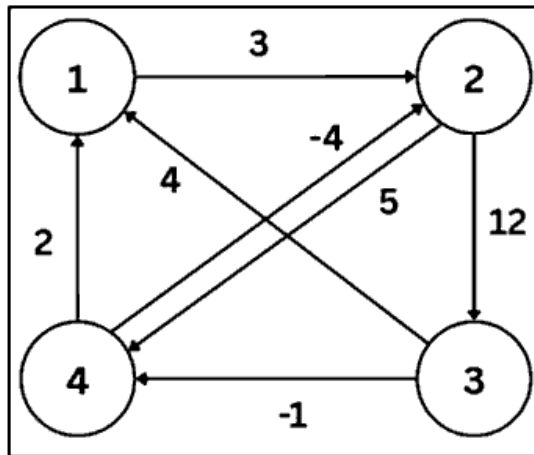


Q4. Solve Any Two of the following.

- A) Obtain the shortest path from source (S) to target (T) of the following multistage graph using backward approach. 6



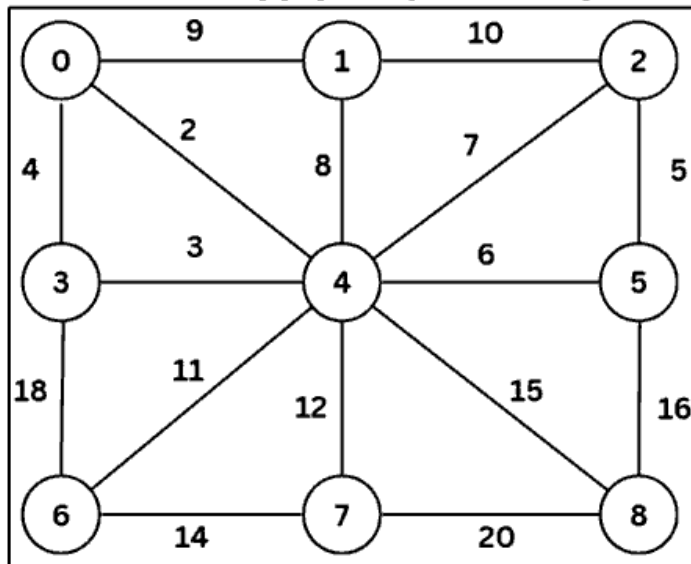
- B) Trace the Floyd-Warshall's algorithm for the following graph and obtain all pair shortest path. 6



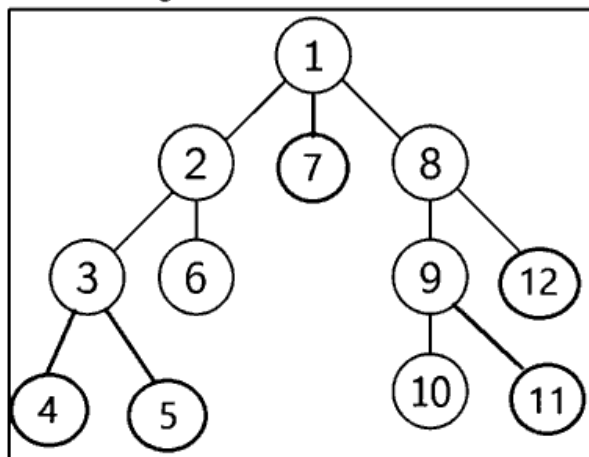
- C) Solve following knapsack instance using dynamic programming : $n=3$, $(w_1, w_2, w_3) = (2, 3, 4)$ and $(p_1, p_2, p_3) = (1, 2, 5)$. Where $M=6$. 6

Q5. Solve Any Two of the following.

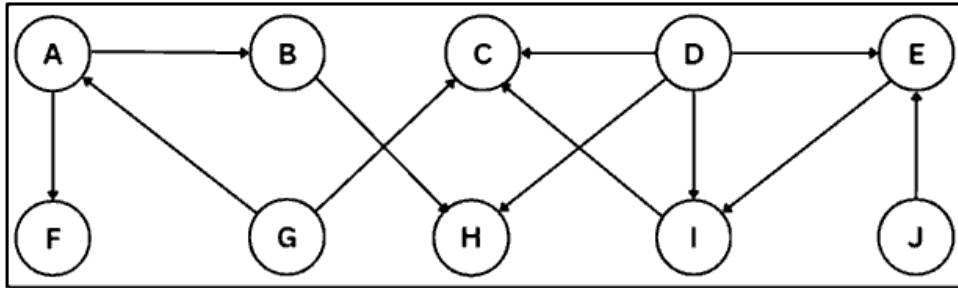
- A) Define Spanning tree, minimum spanning tree. Obtain the minimum cost spanning tree for the following graph using Kruskal's algorithm. 6



- B) Differentiate between BFS and DFS traversal. Write DFS and BFS traversal for the following. 6



- C) Explain topological sort in detail. Apply topological sort and obtain ordering for following graph. 6



Q6. Solve Any Two of the following.

- A) Solve the following sum of subset problem using backtracking and draw a state space tree. $m = 8$ and $w = \{1, 3, 4, 5\}$. 6
- B) Differentiate between backtracking and branch & bound technique. 6
- C) Draw and explain P, NP, NP-Complete and NP-hard Problem. 6

*** End ***